Physics 220 – Practice Final Exam 2

#1 - #15 **[2 points each]** Multiple Choice. Circle the letter of the correct answer.

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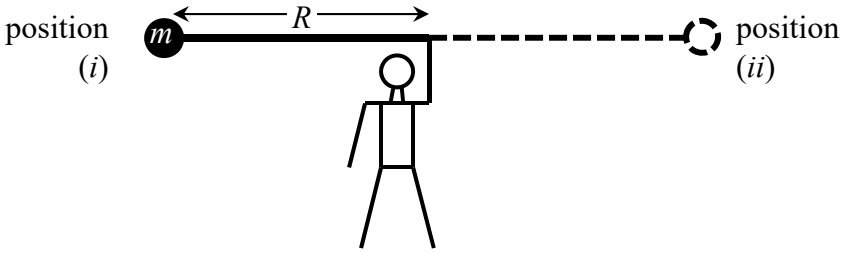
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3) A person holds a light string with a ball of mass *m* attached to the other end. The person whirls the ball in a **horizontal** circle of radius *R*. The person ***gradually*** increases the speed of the ball. When the tension in the string is 100N, the string breaks. What is the kinetic energy of the ball just after the string breaks?
   A) KE = 100R
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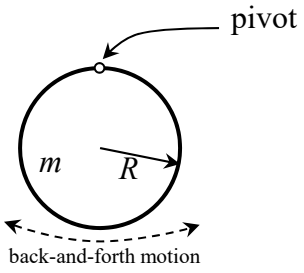
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5) Determine the period of a pendulum that consists of a uniform disk of mass *m* and radius *R*. The disk pivots about a small steel pin through a position on the disk's edge. Assume it pivots with a small angle. For a disk, $I_{c.o.m.} = \frac{1}{2} m R^2$

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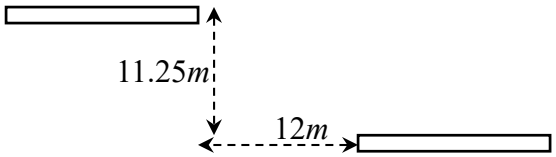
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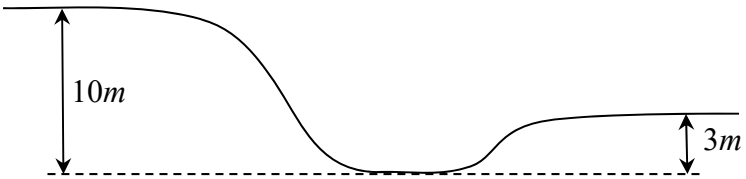
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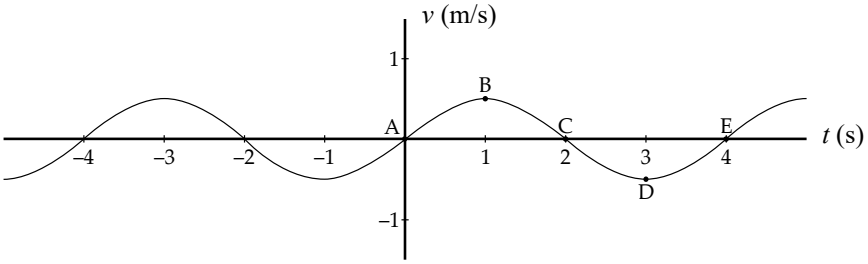


12) A village well in Central Africa is 85.0m deep; *i.e.* The surface of the water is 85m below ground level. When top of the well is uncovered, what is the fundamental resonant frequency of sound in the well? The speed of sound in air is 340 m/s.

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13) A graph of ***velocity versus time*** for an object oscillating at the free end of a horizontal spring is shown below. The point at which the object has zero acceleration and negative velocity is

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14) A string is shaken in such a way that the vertical position of a bit of string at position x and time t is given by:

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What is the speed of the wave traveling down the string?

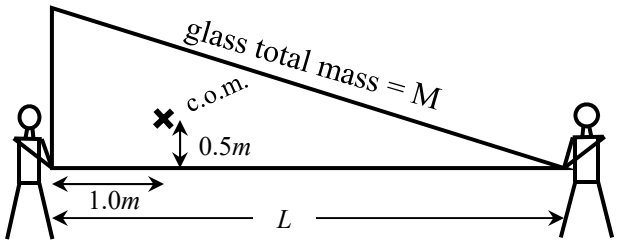
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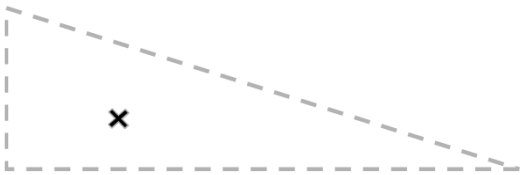
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Physics 220 – Practice Final Exam 2

16) (15 pts) Two workmen lift a big heavy sheet of glass which has a **mass *M***. The glass is triangular in shape, as shown in the figure, with bottom edge of **length *L***. The center of mass of the glass lies 1.0 *meters* from the left workman and 0.5 *meters* above the bottom edge of the glass. The workmen hold the glass stationary with the bottom edge horizontal. The left workman exerts force F_L and the right workman exerts force F_R as shown. Working as a team, *they each exert the minimum force possible* to keep the glass stationary. Use g = 10 m/s/s.



A) (3 pts) Draw the extended free body diagram showing the *direction and correct locations* of the forces that act on the sheet of glass.



B) (8 pts) Determine F_L and F_R , the forces that the workmen exert. Your answers should be written in terms of the length of the glass, *L*, the mass of the glass, *M*, and g. *Hint: Choose the left workman's hands as the pivot.*

F_L =

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C) (4 pts) In terms of *M*, the moment of inertia of the glass about its center-of-mass is $I_{c.o.m.} = \mathbf{5M}$ [kg•m^2]. The right workman accidentally drops his end of the glass, but the left workman still exerts the same force F_L which holds his corner fixed. What is the resulting angular acceleration, α, of the glass the instant the right end is dropped?

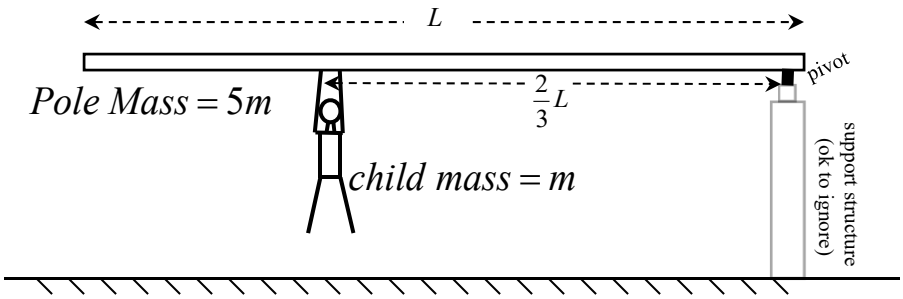
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General : for a stick of mass M, length ℓ

$I_{\substack{stick \\ c.o.m.}} = \frac{1}{12} M\ell^2$



2 pts each; fill in the boxes below, your answers should be in terms of *m* and *L*

Before child grabs pole

$I_{\substack{pole \\ pivot}} = \frac{5}{3} mL^2$ ← Note: The moment of inertia of this pole by itself around its end is already given

$\omega_i = 1.0 \frac{rad}{s}$

$L_i =$ _____

$K.E._i =$ _____

After child grabs pole

$I_{\substack{system \\ about \\ pivot}} =$ _____

$\omega_f =$ _____

$L_f =$ _____

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(3 pts): Show that the moment of inertia of this pole by itself about the pivot is $I_{end} = (5/3)\, m\, L^2$;
Start with $I_{\substack{stick \\ c.o.m.}} = \frac{1}{12} M\ell^2$ Show all work!

Physics 220 – Practice Final Exam 2

18) (10 pts) An object moves with simple harmonic motion according to the following equation:

$$x(t) = 6.0(meters)\cos(3\pi(rad/s)t + {}^{\pi}/_{3}(rad))$$

Fill out the values below (1 pt each):

$T =$ _____

$v_{max} =$ _____

Amplitude = _____

$\omega =$ _____

$f =$ _____

$a_{max} =$ _____

Total Distance The Object Travels in One Period = _____

At **$t = 2sec$** what are the following values?

displacement from equilibrium = _____

velocity = _____

acceleration = _____

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19) A) (6 points) Draw the first 3 resonant modes for a guitar string below. When the string is at rest, it lies along the dotted line:



Fundamental mode (1st Harmonic)

2nd Harmonic

3rd Harmonic

B) (4 pts) The guitar string is 1.0 *meter* long and the fundamental mode has a frequency $f_1 = 100$ Hz. If the string has a mass of 50 grams, what is the tension in the string?

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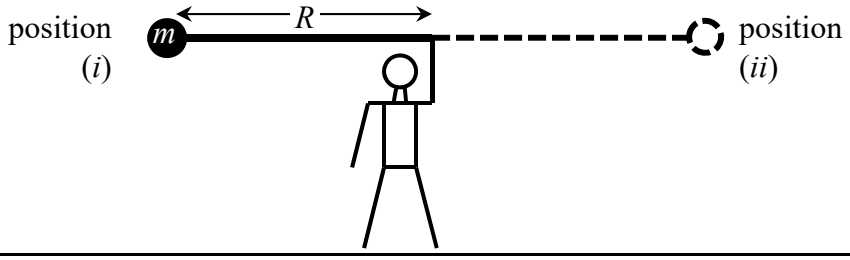
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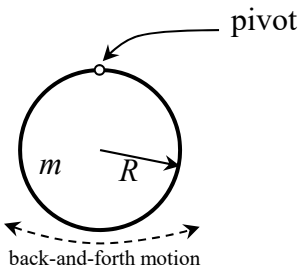
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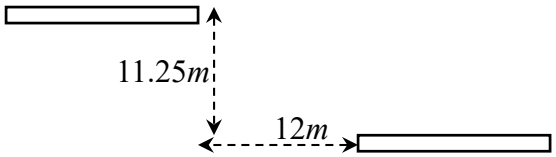
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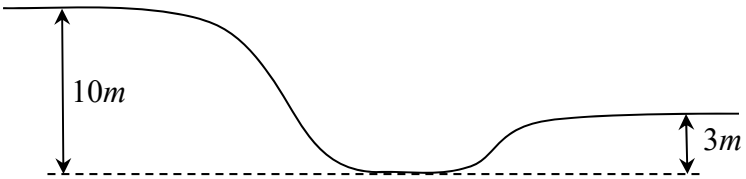
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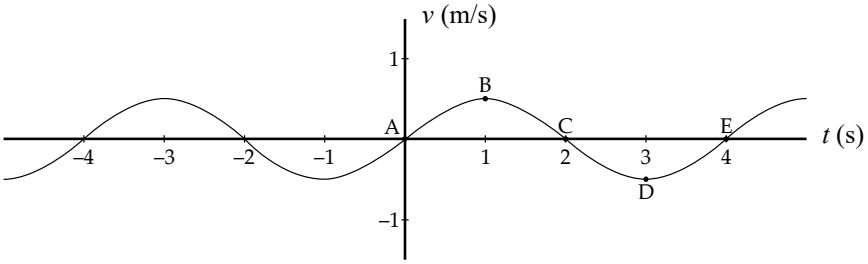


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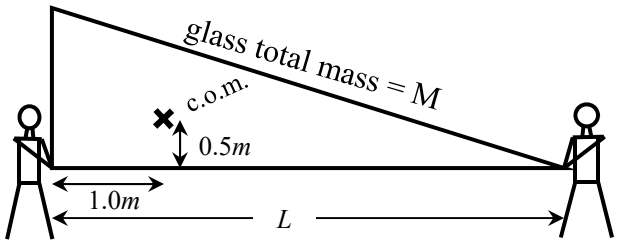
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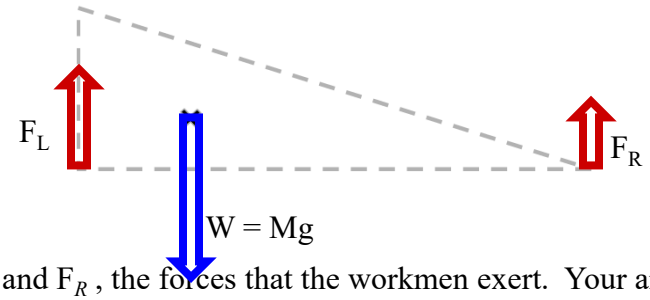
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$$\tau_{net} = I\alpha = 0$$

$$\tau_{\substack{into\\paper}} = r_\perp F = (1.0)Mg$$

$$\tau_{\substack{out\ of\\paper}} = rF_\perp = LF_R$$

$$\Rightarrow 1Mg = LF_R$$

$$F_R = \frac{Mg}{L}$$

$$F_{net} = ma = 0$$

$$F_R + F_L - W = 0$$

$$\frac{Mg}{L} + F_L = Mg$$

$$F_L = Mg\left(1 - \frac{1}{L}\right)$$

$$F_L = Mg\left(1 - \frac{1}{L}\right)$$

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C) (4 pts) In terms of *M*, the moment of inertia of the glass about its center-of-mass is $I_{c.o.m.} = \mathbf{5M}$ [kg•m^2]. The right workman accidentally drops his end of the glass, but the left workman still exerts the same force F_L which holds his corner fixed. What is the resulting angular acceleration, α, of the glass the instant the right end is dropped?

$$I_{c.o.m.} = 5M$$

$$I_{\substack{left\\corner}} = 5M + M(\sqrt{1^2 + 0.5^2})^2$$

$$I_{\substack{left\\corner}} = 6.25M$$

$$\tau_{net} = (1.0)Mg = I_{\substack{left\\corner}}\alpha$$

$$\Rightarrow \alpha = \frac{Mg}{6.25M} = \frac{g}{6.25} \approx 1.6\ \frac{rad}{s^2}$$

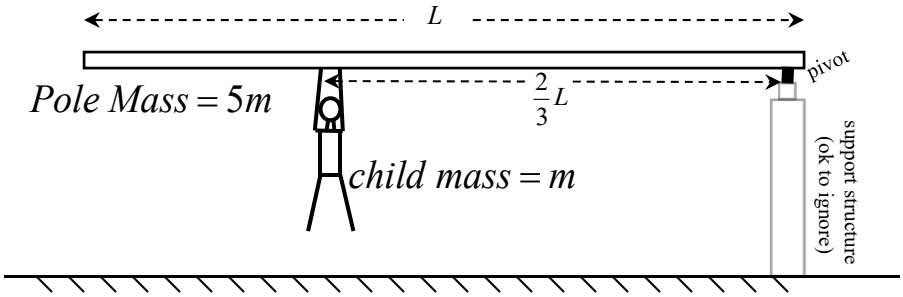
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$\omega_i = 1.0 \frac{rad}{s}$

$L_i =$ _____ $\frac{5}{3} mL^2$

$K.E._i =$ _____ $\frac{5}{6} mL^2$

After child grabs pole

$I_{\substack{system\\about\\pivot}} =$ _____ $\frac{19}{9} mL^2$

$\omega_f =$ _____ $\frac{15}{19} \frac{rad}{s}$

$L_f =$ _____ $\frac{5}{3} mL^2$

$K.E._f =$ _____ $\frac{25}{38} mL^2$

initial

$L_i = I_i\omega_i = \frac{5}{3} mL^2 (1.0 \frac{rad}{s}) = \frac{5}{3} mL^2$

$K.E._i = \frac{1}{2} I_i\omega_i^2 = \frac{1}{2} (\frac{5}{3} mL^2)(1.0)^2 = \frac{5}{6} mL^2$

final

$L_f = I_f\omega_f = L_i$

$\Rightarrow (I_{rod} + I_{child})\omega_f = I_i\omega_i$

$((\frac{5}{3} mL^2) + m(\frac{2}{3} L)^2)\omega_f = \frac{5}{3} mL^2$

$mL^2(\frac{5}{3} + \frac{4}{9})\omega_f = \frac{5}{3} mL^2$

$\omega_f = \frac{\frac{5}{3}}{\frac{15}{9} + \frac{4}{9}} = \frac{5}{3}\frac{9}{19} = \frac{15}{19} \frac{rad}{s}$

$K.E._f = \frac{1}{2} I_f\omega_f^2 = \frac{1}{2} (\frac{19}{9} mL^2)(\frac{15}{19})^2 = \frac{1}{2}\frac{25}{19} mL^2 = \frac{25}{38} mL^2$

$\Rightarrow$ *less than initial energy*!

(3 pts): Show that the moment of inertia of this pole by itself about the pivot is $I_{end} = (5/3)\, m\, L^2$;
Start with $I_{\substack{stick\\c.o.m.}} = \frac{1}{12} M\ell^2$ Show all work!

$I_{c.o.m.} = \frac{1}{12} M\ell^2$

$I_{end} = I_{c.o.m.} + (5m)(\frac{L}{2})^2$

$I_{end} = \frac{1}{12}(5m)L^2 + \frac{5}{4} mL^2$

$I_{end} = \frac{5+15}{12} mL^2 = \frac{20}{12} mL^2$

$I_{end} = \frac{5}{3} mL^2$

Physics 220 – Practice Final Exam 2

18) (10 pts) An object moves with simple harmonic motion according to the following equation:

$$x(t) = 6.0(meters)\cos(3\pi(rad/s)t + \pi/3(rad))$$

Fill out the values below (1 pt each):

$T =$ ___$\frac{2}{3}s$___

$v_{max} =$ ___$18\pi \frac{m}{s}$___

$Amplitude =$ ___$6.0meters$___

$\omega =$ ___$3\pi \frac{rad}{s}$___

$f =$ ___$\frac{3}{2}Hz$___

$a_{max} =$ ___$54\pi^2 \frac{m}{s^2}$___

Total Distance The Object Travels in One Period = ___$24meters$___

At ***t* = 2sec** what are the following values?

displacement from equilibrium = ___$3\sqrt{3}meters$___

velocity = ___$-9\pi \frac{m}{s}$___

acceleration = ___$-(27)(\sqrt{3})\pi^2 \frac{m}{s^2}$___

by inspection :

$\omega = 3\pi$

$Amplitude = 6.0meters$

$\phi = \frac{\pi}{3}$

Then;

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3\pi} = \frac{2}{3}$$

$$f = \frac{1}{T} = \frac{3}{2}$$

$$v_{max} = A\omega = 6(3\pi) = 18\pi$$

$$a_{max} = A\omega^2 = 6(3\pi)^2 = 54\pi^2$$

$$Tot.Dist.Traveled = 4(Amplitude) = 4(6) = 24meters$$

$$x(2) = 6\cos(3\pi(2) + \tfrac{\pi}{3})$$

$$x(2) = 6\cos\tfrac{\pi}{3} = 6\tfrac{\sqrt{3}}{2} = 3\sqrt{3}$$

$$\frac{dx(t)}{dt} = v(t) = -6(3\pi)\sin(3\pi t + \tfrac{\pi}{3})$$

$$v(t) = -18\pi\sin(3\pi t + \tfrac{\pi}{3})$$

$$v(2) = -18\pi\sin(3\pi(2) + \tfrac{\pi}{3})$$

$$v(2) = -18\pi(\tfrac{1}{2}) = -9\pi$$

$$\frac{dv(t)}{dt} = a(t) = -18\pi(3\pi)\cos(3\pi t + \tfrac{\pi}{3})$$

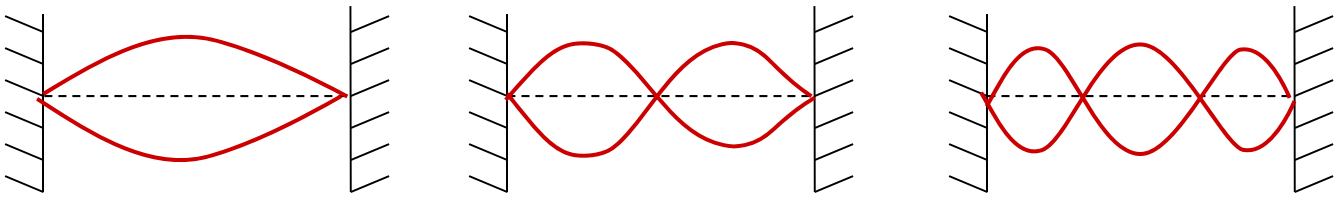
$$a(t) = -54\pi^2\cos(3\pi t + \tfrac{\pi}{3})$$

$$a(2) = -54\pi^2\cos(3\pi(2) + \tfrac{\pi}{3})$$

$$a(2) = -54\pi^2\tfrac{\sqrt{3}}{2} = -27\sqrt{3}\pi^2 = -46.8\pi^2$$

Physics 220 – Practice Final Exam 2

19) A) (6 points) Draw the first 3 resonant modes for a guitar string below. When the string is at rest, it lies along the dotted line:



Fundamental mode (1st Harmonic)

2nd Harmonic

3rd Harmonic

B) (4 pts) The guitar string is 1.0 *meter* long and the fundamental mode has a frequency f_1 = 100 Hz. If the string has a mass of 50 grams, what is the tension in the string?

$$L = 1.0m$$
$$\lambda_1 = 2L = 2.0m$$
$$f_1 = 100Hz$$
$$m = 0.050kg$$
$$\mu = \frac{m}{L} = \frac{0.050}{1.0} = 0.050$$

$$\sqrt{\frac{Tension}{\mu}} = v_{\substack{sound \\ in\ wire}} = \lambda f$$

$$Tension = \lambda^2 f^2 \mu = (2.0)^2 (100)^2 (0.050)$$

$$\boxed{Tension = 2,000\,Newtons}$$